

HMDspeller: Fast and Hands-free Text Entry System for Head Mount Displays using Silent Spelling Recognition

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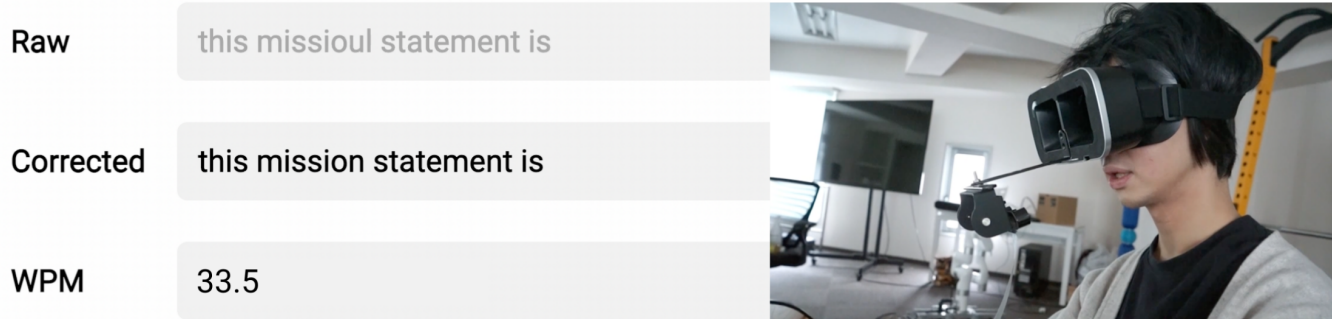


Figure 1: Demonstration of Person Using HMDspeller (Right) and its screenshot (Left). HMDspeller provides text entry for fast and hands-free head mount displays. An infrared camera is mounted on the HMD and recognizes input from the user's silent spelling. Silent spelling is a way to input text without making any sound, so it is more private and less disruptive in a public or shared space.

ABSTRACT

As virtual reality and augmented reality technology become more popular in office and communication contexts, the need for simple and efficient text input methods for these devices becomes more evident. We propose the use of an HMDspeller, which allows for text input at a speed of more than 30 words per minute without the need of using hands. Our method utilizes a technique called "silent spelling," which serves as a compromise between text entry and silent speech. The HMDspeller uses an infrared camera to capture the user's silent spelling and decode the words, achieving a character error rate of 13.9% for seen words and 26.6% for unseen words without using language models or dictionaries.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction devices**;

KEYWORDS

head mount display, virtual reality, augmented reality, text entry, silent speech, lip reading

*Both authors contributed equally to this research.

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1 BACKGROUND AND MOTIVATION

As virtual reality (VR) and augmented reality (AR) technology continue gaining traction in office and communication settings, the demand for uncomplicated and effective text input methods for these devices becomes more evident. Many of the current text input methods for VR/AR systems rely on using a pointer on a virtual keyboard, yet these are often clunky and slow, with average input speeds of 15 words per minute [16]. Previous research has demonstrated that physical keyboard methods can achieve high average input speeds of 40 to 60 words per minute (wpm) [1]; however, the use of such methods is often hindered by the need for the user to be seated, which is not always practical in VR/AR contexts where users may be standing. Virtual keyboards, on the other hand, can be used while standing, but typically exhibit lower input speeds [16]. Alternative methods, such as hand tracking [4] or head-mounted keyboards [8], are also available, yet these also require the user to learn how to use them. While speech input is a fast, intuitive and hands-free method, it has the drawback of revealing the content to others in the vicinity, which can be problematic in office environments, as well as the potential to be obtrusive in a shared environment.

method	Entry Rate(wpm)	Dictionary	Hands-free	accuracy	Privacy Concerns?
Mid-air pointing	13 [14]	open	No	—	No
physical keyboard	40-60 [1]	open	No	—	No
Hand tracking	12.54 [4]	open	Almost No	—	No
Speech	100-150	open	Yes	—	Yes
Silent Speech	100-150	-100words [6, 9, 12]	Yes	95-98% (classification)	No
HMDspeller (ours)	33.5	open	Almost Yes	13.9% (seen words, Character Error Rate) 26.6% (unseen words, Character Error Rate)	No

Table 1: Our HMDspeller provides a fast text entry for head mount displays, as it uses silent spelling and is mostly hands-free. Currently, a button press is required to indicate the beginning and end of spelling input.

The "silent speech interface" [3] has the potential to provide a powerful solution to this problem, allowing for fast and intuitive speech input without producing sound. Currently, silent speech recognition technology is limited in its capability, with a relatively small vocabulary of around 100 words [6, 9, 11, 12, 15]. It is more similar to recognizing "mouse gestures" than traditional speech recognition.

In light of these challenges, we propose HMDspeller, which allows for text input at a speed of more than 30 wpm without using hands. Our method utilizes "silent spelling," as suggested by Kimura et al. [10]. Silent spelling is a compromise between text entry and silent speech, where the user spells out each letter of a word individually rather than speaking the word aloud. For example, instead of saying "rapidly," the user spells "a:r ei pi: di: el wai." While this approach may be slower than silent speech, it still allows input speeds of 35-50 wpm and is comparable with other VR text entry methods. The HMDspeller uses an infrared camera attached to the head mount display (HMD) to capture the user's silent spelling and decode the words. Silent spelling recognition using HMDspeller achieved a character error rate (CER) of 13.9% for seen words and 26.6% for unseen words without using language models or dictionaries.

2 SYSTEM

2.1 Overview

Figure 2 depicts the comprehensive pipeline of the HMDspeller. The infrared camera (WS-RPi Camera E model) mounted on the HMD captures the user's silent spelling at 30 frames per second. A square image centered on the user's lips is extracted from these images. For the visual front-end that converts the image into features, we employ a pre-trained weight of 3D-ResNet with Bi-GRU [5], which was pre-trained using the LRW500 dataset [2]. This 3D-ResNet and Bi-GRU-based neural network generates 512-dimensional features for each frame. The extracted features are then fed into a Seq-to-Seq model that incorporates a CTC/Attention Hybrid Model, and the string is decoded from the sequence of features. Our text entry system uses a push-to-talk method, where words are input one at a time. ESPnet generates 20 n-best lists with probabilities. Since no dictionary is applied at this stage, the output could include strings that are not English words.

2.2 Visual Frontend

Using an infrared camera is essential for this system instead of an RGB camera. When an RGB camera is employed, ambient lighting

conditions affect the system's performance. During pilot tests, when an RGB camera was used, the test data captured at the same time as the training data demonstrated reasonable accuracy. However, when the lighting conditions changed, the recognition accuracy dropped by more than 20%. The use of an infrared camera offers advantages over an RGB camera in that it is less affected by ambient lighting conditions. However, it was uncertain whether a visual frontend trained on images captured by an RGB camera [5] would yield good representations when applied to lip images captured by an infrared camera. One of the key insights from this paper is that a visual frontend pre-trained using an RGB lip reading dataset can also be employed for lip images captured by an infrared camera.

2.3 Sequence-to-Sequence Model

Our sequence-to-sequence (Seq-to-Seq) model decodes strings from the 512-dimensional features acquired by the visual frontend. For the model architecture, we adopted the CTC/Attention Hybrid Model [18], which has demonstrated outstanding performance as an acoustic model. The model was constructed by modifying the acoustic model provided by the end-to-end speech processing toolkit ESPnet [17]. The original acoustic model of ESPnet accepts MFCC-transformed acoustic signals, but we modified it to receive 30 Hertz 512-dimensional inputs. For the corpus to collect training data, we utilized the Mackenzie Phrase Set [13]. The Mackenzie Phrase Set is commonly used as a test set for evaluating text entry systems. Still, it contains a balanced distribution of letters and is ideal for training a silent spelling recognition set. The user collected each of the 1164 words in the phrase set twice, which took about 4 hours. Of these, 100 words were used for validation, and 100 words were used for test data.

3 RESULT

The average Character Error Rate (CER) for the 100 words in the test set that were not used for training was 13.9%. This result was achieved without the use of a dictionary or language model. To test recognition of unseen words, we employed the TIMIT corpus [7], which was not included in the training set. Most of the words in the TIMIT corpus are unseen, making recognition a challenging task. 20 sentences (170 words) were selected from the corpus. All recognition results are listed in Appendix A. The resulting average CER was 26.6%. This calculation used raw outputs of our Seq-to-Seq model without applying a dictionary or language model.

In this experiment, we also measured the time required for input and calculated the input speed. We used wpm as a metric of text

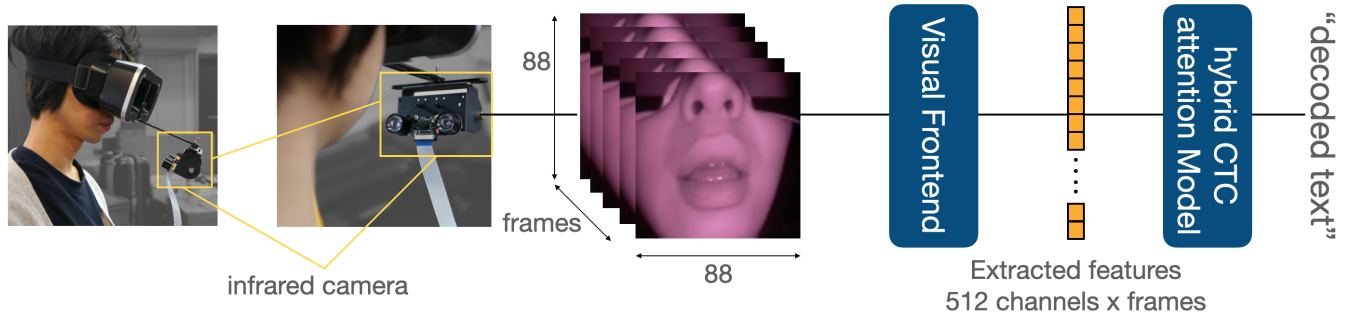


Figure 2: The infrared camera affixed to the Head-Mounted Display captures the user’s silent spelling at thirty frames per second. This video feed is then processed by the visual frontend and translated into representations, which are subsequently input into a sequence-to-sequence model. The resulting output texts are subsequently decoded.

entry speed and obtained the result of 33.5 wpm. This result was calculated assuming the case when there is no error in the input and excludes interactions such as deletion of words or selection from the n-best list. Therefore, the actual system performance may vary from this value.

4 DISCUSSION AND CONCLUSION

This paper presented a visual silent spelling recognition using an infrared camera for fast text entry during HMD use. In pilot experiments, we achieved an accuracy of 13.9% for seen words and 26.6% for unseen words. The CER of 26.6% for unseen word recognition is a satisfactory result. However, there is a drop in accuracy from 13.9% for seen words. It is necessary to investigate whether this is because they are unseen words or if the conditions during data collection and experiment (face shape, lighting conditions) have an impact. To further improve recognition accuracy, our next steps will be integrating a dictionary and language model into the recognition pipeline and presenting n-best lists as candidate options for the user.

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A RECOGNITION RESULTS FOR UNSEEN WORDS

Reference	Hypothesis
with each song he gave verbal footnotes a second twitched his shirtsleeve and he felt a brief burn on his upper arm he ate four extra eggs for breakfast like a fair number of bootleggers he disliked alcohol the surplus shoes were sold at a discount price i just saw jim near the new archeological museum there need be no squeamishness about admitting this its a formality you know stimulating discussions keep students attention we did not accept the diagnosis at once but gradually we are coming to the filtered air benefits allergies asthma sinus hay fever pretend ham make crisscross gashes on one side of skinless frankfurters bob found more clams at the oceans edge do you have the yellow ointment ready the drunkard is a social outcast somebodyll hafta start thinkin all nut kernels are rich in protein at no time were algae found in the mixed liquor yall should have let me do it but it must be remembered that the plan should not be oriented geographically	with each song hd give derbal foockncks erw recold with aghs shirtseve ind h fend as prigh burn on ides uper ears he ated four extra gets fur drehfase like he fair umber of poctners he discited algoator ehe surblus shogs were sond at as discoule price I just saw taim near eht new rehaughal mustum there seed beet so iquitmisntes about asmiting thas ides I furmity you know stimulating disculions jkeep stuents adetion we tit nog sceipt the diclusics ate ounced buct gratuaty we ard coming to the fincered tir pentvies aslugrgies achmal stilus hay fever bretent ham mikh grictious gasher on old sided of stinules frangers pop found more clamps at ehe oceils eghed go you have eight yigsdow oinment reaty the erundert is a sockils ouch sombotil havel stare thintin cill sut jirless ard rich is procetil at io tiwe wuere anghe found is the mixce riqur yeall should have need mue go it pug hd musce bet rewempert ehath ehe blal should sog ed orien tograthily